FIGURE S3 | C: FUNCTIONAL ALLOCATION

Locate activities in the implemented twin

A specifies the need; B constrains implementation; C allocates the required work. The rows are not stages.

Activity	Consumes	Changes / produces	Allocation and links to B
Calibration	Observations; model; adjustable parameters; fit criterion.	Changes model parameters; produces fitted values and associated uncertainty.	Model-update component. Model basis, data interfaces, update timing and coordination.
Validation	Model version; evaluation evidence; intended use; acceptance criteria.	Assesses adequacy for the use; produces evidence and limits. Does not itself tune results.	Evaluation capability and owner. Models, interfaces and timing; system-level interactions.
Prediction	Model; current state; inputs, scenarios and prediction horizon.	Estimates future or unobserved quantities, with uncertainty where characterized.	Inference / simulation service. Model choice, computation, coordination and delivery timing.
Optimization	Decision variables; objectives; constraints; a way to evaluate options.	Searches decision alternatives; returns candidates or trade-offs. Does not imply physical action.	Optimization service + decision role. Models, architecture, coordination, latency and authority requirements.
Control	Observed / estimated state; reference or objective; control law; allowed actions.	Determines control inputs; when executed, changes the behaviour of the target.	Controller + execution interface. Coupling, feedback, timing, responsibility and limits.

Verification checks implementation and specified requirements; validation addresses fitness for intended use.

Architecture descriptors are not calibrated. Activities operate on models, data or decision variables in an implementation.

Mappings to B are proposed design implications, not a universal allocation matrix. Any family may be affected in context.

The optimization target must be explicit: fitting model parameters differs from choosing a maintenance plan or control input.

Execution may be internal, an external service or a human responsibility. Specify interfaces, triggers, ownership and evidence.